APPENDIX A

To provide a clearer depiction of the QTCO-GBC method, Algorithm A1 illustrates complete procedural framework.

Algorithm A1: The QTCO-GBC framework

Input: Datasets
Parameter: α, B, τ_h, υ_h, ς
Output: The ranking results **Step 1. MDCM-AAE**
while $k \leq q$ **do**
  Encode X^k at fine, medium, and coarse scales.
  Decode $Z^{k,\text{fine}}$, $Z^{k,\text{mid}}$ and $Z^{k,\text{coarse}}$ to form multi-scale decision matrix.
  Learn scale weights to obtain single-scale information.
end
Step 2. GBG-IJGP
foreach $SG^h \in Sub$ **do**
  while $i \leq m$ *and* $j \leq n$ **do**
    Construct the GB GB_{ij}^h based on objects and attributes.
    Compute coverage $\text{Cov}(GB_{ij}^h)$, specificity $\text{Spe}(GB_{ij}^h)$, and entropy H_{ij}^h.
  end
  Select the optimal GB.
  Compute the subgroup quality M^h and weight w^h.
end
Aggregate subgroup GBs into group structure.
Step 3. BTOM-QFOM
foreach $SG^h \in Sub$ **do**
  Obtain the subgroup consensus Con^h.
end
Obtain the global consensus Gcon.
Generate the most stable trust relationships T^* based on Model (M-1).
while $Gcon < \varsigma$ **do**
  Adjust information base on Model (M-2).
end
Use Model (M-3) and the optimized information to compute the quantum probability of objects.
Rank objects based on quantum probability to obtain the final decision results.
return *solution*

According to Algorithm A1, the time complexity of Step 1 is $O(3qmn)$, Step 2 is $O(N^2mn)$, and Step 3 is $O(q^2mn)$. Since Step 3 dominates, the overall time complexity of the algorithm is $O(q^2mn)$.

APPENDIX B

Thirty normal perturbations with a mean of 0 and a standard deviation of 0.1 are generated and combined with the original data to construct 30 related datasets, which are regarded as the evaluation opinions of DAs. Subsequently, based on MDCM-AAE, the initial data of each DA are converted into multi-scale data, as shown in Table B1, and then fused into single-scale data through scale weight learning. It is worth noting that Table B1 presents an example of the decision information of one DA.

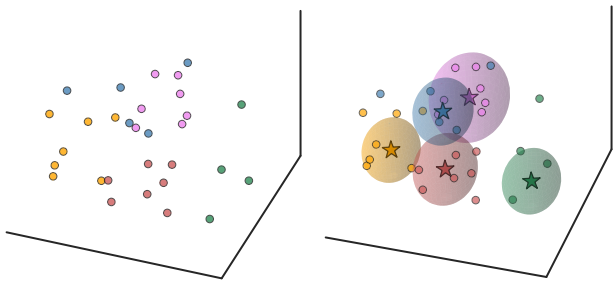
Based on GBG-IJGP, the clustering results of the group and their corresponding generated GB structures are first determined, as shown in Fig. B1. Then, using the generated GB structures, the quality and weights of each subgroup are calculated, as presented in Table B2. The group quality is subsequently determined, with the resulting value being 0.771.

According to the consensus measurement method, the consensus degrees of the subgroups are 0.938, 0.947, 0.901, 0.944, and 0.858, respectively, while the group consensus is 0.931, which is below the consensus threshold of 0.950. Therefore, BTOM is first applied to obtain the most stable trust relationships among subgroups. Then, OFOM is used for consensus adjustment. After 0.371 seconds of iterations, overall consensus is achieved, with the group consensus increasing to 0.959. Finally, based on Model (M-3), the final ranking of objects is obtained as follows: $u_9 \succ u_8 \succ u_2 \succ u_4 \succ u_{10} \succ u_3 \succ u_1 \succ u_6 \succ u_7 \succ u_5$.

In summary, the case analysis in this section demonstrates that the proposed method can effectively reconcile differences among DAs in a short period of time, achieving a high level of group consensus and further validating its effectiveness and applicability.

TABLE B1: Multi-scale data.

	a_1^1	a_1^2	a_1^3	a_2^1	a_2^2	a_2^3	a_3^1	a_3^2	a_3^3	a_4^1	a_4^2	a_4^3	a_5^1	a_5^2	a_5^3	a_6^1	a_6^2	a_6^3
u_1	0.710	0.721	0.721	0.248	0.213	0.213	0.730	0.826	0.826	0.116	0.102	0.102	0.637	0.532	0.532	0.548	0.671	0.671
u_2	0.628	0.645	0.646	0.343	0.303	0.338	0.275	0.407	0.405	0.177	0.167	0.175	0.746	0.622	0.719	0.212	0.314	0.314
u_3	0.739	0.721	0.721	0.139	0.213	0.213	0.907	0.826	0.826	0.073	0.102	0.102	0.345	0.532	0.532	0.776	0.671	0.671
u_4	0.681	0.669	0.646	0.278	0.278	0.338	0.401	0.529	0.405	0.125	0.137	0.175	0.756	0.712	0.719	0.255	0.342	0.314
u_5	0.562	0.562	0.562	0.906	0.906	0.906	0.098	0.098	0.098	0.779	0.779	0.779	0.917	0.917	0.917	0.731	0.731	0.731
u_6	0.696	0.669	0.646	0.219	0.278	0.338	0.607	0.529	0.405	0.102	0.137	0.175	0.692	0.712	0.719	0.352	0.342	0.314
u_7	0.713	0.721	0.721	0.266	0.213	0.213	0.799	0.826	0.826	0.125	0.102	0.102	0.606	0.532	0.532	0.683	0.671	0.671
u_8	0.658	0.645	0.646	0.273	0.303	0.338	0.541	0.407	0.405	0.160	0.167	0.175	0.488	0.622	0.719	0.439	0.314	0.314
u_9	0.607	0.607	0.646	0.612	0.612	0.338	0.129	0.129	0.405	0.336	0.336	0.175	0.856	0.856	0.719	0.281	0.281	0.314
u_{10}	0.626	0.669	0.646	0.351	0.278	0.338	0.569	0.529	0.405	0.202	0.137	0.175	0.690	0.712	0.719	0.436	0.342	0.314



(a) (b)

Fig. B1: Generated GB structures.

TABLE B2: Quality and weight results.

	SG^1	SG^2	SG^3	SG^4	SG^5
Quality	0.639	0.702	0.584	0.701	0.490
Weights	0.247	0.271	0.135	0.271	0.076